

Activity Sheet – 6

Dictionaries

Objectives:

1. Understand how dictionaries store key-value pairs and their use in data organization.
2. Create, access, modify, and delete dictionary entries using built-in methods.
3. Apply dictionaries in solving real-world mapping and counting problems efficiently.

Reading:

1. Definition and Initialization
`student = {"name": "John", "age": 20, "grade": "A"}`
2. Accessing Values
 - Using keys: `student["name"]`
 - Using `.get()`: `student.get("marks", "Not Found")`
3. Modifying Values
 - Update: `student["grade"] = "A+"`
 - Add new key-value pair: `student["id"] = 101`
4. Dictionary Methods
 - `.keys()`, `.values()`, `.items()`, `.update()`, `.pop()`, `.clear()`, `del dict[key]`
5. Looping through Dictionary
`for key, value in student.items():`
`print(key, ":", value)`
6. Dictionary Comprehension
`squares = {x: x**2 for x in range(5)}`
7. Use Cases
 - Frequency counting
 - Mapping roll numbers to names
 - Storing structured data efficiently

Illustrative Examples:

Example #1:

Problem Statement: Write a Python program to create a dictionary for a student with name, age, and grade. Display all the details.

Python Code:

```
student = {"name": "Alice", "age": 19, "grade": "A"}
print(student)
```

Sample Input:

Name: Alice

Age: 19

Grade: A

Sample Output:

```
{'name': 'Alice', 'age': 19, 'grade': 'A'}
```

Example #2:

Problem Statement: Write a program to print the grade of a student from a dictionary.

Python Code:

```
student = {"name": "Alice", "age": 19, "grade": "A"}  
print("Grade:", student["grade"])
```

Sample Output:

Grade: A

Programming Questions:

Exercise: 1

Difficulty: Easy

Question: Write a Python program to merge two dictionaries into one.

Program

```
dict1 = {'a': 1, 'b': 2, 'c': 3}
dict2 = {'d': 4, 'e': 5, 'c': 6}
merged_dict = {}
for x in dict1:
    merged_dict[x] = dict1[x]
for y in dict2:
    merged_dict[y] = dict2[y]
print("Merged Dictionary:", merged_dict)
```

Output:

```
Merged Dictionary: {'a': 1, 'b': 2, 'c': 6, 'd': 4, 'e': 5}
```

Exercise: 2

Difficulty: Easy

Question: Write a Python program to create a dictionary with integers as keys and their squares as values (e.g., {1: 1, 2: 4, 3: 9}).

Program

```
n = 25
square_dict = {}
for i in range(1, n + 1):
    square_dict[i] = i * i
print(square_dict)
```

Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64, 9: 81, 10: 100, 11: 121, 12: 144, 13: 169, 14: 196, 15: 225, 16: 256, 17: 289, 18: 324, 19: 361, 20: 400, 21: 441, 22: 484, 23: 529, 24: 576, 25: 625}
```

Exercise: 3

Difficulty: Medium

Question: Write a Python program to count the number of vowels in each word of a sentence and store them in a dictionary.

Program

```
sentence = input("Enter a sentence: ")
vowels = "aeiouAEIOU"
words = sentence.split()
vowel_counts = {}
for word in words:
    count = 0
    for char in word:
        if char in vowels:
            count += 1
    vowel_counts[word] = count
print("Vowel counts in each word:", vowel_counts)
```

Output:

```
Enter a sentence: count the number of vowel in each word
Vowel counts in each word: {'count': 2, 'the': 1, 'number': 2, 'of': 1, 'vowel': 2, 'in': 1, 'each': 2, 'word': 1}
```

Exercise: 4

Difficulty: Medium

Question: Write a Python program to input student marks and find the average, then print those who scored above average.

Program

```
num_students = int(input("Enter the number of students: "))
marks = {}
for i in range(num_students):
    name = input("Enter student name: ")
    mark = float(input("Enter marks: "))
    marks[name] = mark
total = 0
for mark in marks.values():
    total += mark
average = total / num_students
print(f"Average marks: {average:.2f}")
print("Students who scored above average:")
for name, mark in marks.items():
    if mark > average:
        print(name, mark)
```

Output:

```
Enter the number of students: 3
Enter student name: Rudresh
Enter marks: 72
Enter student name: Aakash
Enter marks: 69
Enter student name: Dhivas
Enter marks: 66
Average marks: 69.00
Students who scored above average:
Rudresh 72.0
```

Exercise: 5

Difficulty: Hard

Question: Write a Python program to count word frequency in a paragraph, ignoring punctuation and case.

Program

```
paragraph = input("Enter a paragraph: ")
paragraph = paragraph.lower()
punctuations = "[](){};:'\"\\, <> . / ? @ # $ % ^ & * ~ ! ` "
clean_paragraph = ""
for char in paragraph:
    if char not in punctuations:
        clean_paragraph += char
words = clean_paragraph.split()
word_freq = {}
for word in words:
    if word in word_freq:
        word_freq[word] += 1
    else:
        word_freq[word] = 1
print("Word Frequencies:")
for word, count in word_freq.items():
    print(f"{word}: {count}")
```

Output

```
Enter a paragraph: Python dictionaries are a type of data structure that store data in key-value pairs. Each key in a dictionary is unique, and it is used to access its corresponding value. Dictionaries are mutable, which means you can add, remove, or change items. They are widely used in Python programming for storing and organizing data efficiently
Word Frequencies:
python: 2
dictionaries: 2
are: 3
a: 2
type: 1
of: 1
data: 3
structure: 1
that: 1
store: 1
in: 3
key-value: 1
pairs: 1
each: 1
key: 1
dictionary: 1
is: 2
unique: 1
and: 2
it: 1
used: 2
to: 1
access: 1
its: 1
corresponding: 1
value: 1
mutable: 1
which: 1
means: 1
you: 1
can: 1
add: 1
remove: 1
or: 1
change: 1
items: 1
they: 1
widely: 1
programming: 1
for: 1
storing: 1
organizing: 1
efficiently: 1
```

Exercise: 6

Difficulty: Hard

Write a Python program to create a dictionary from a string. Note: Track the count of the letters from the string. String: 'w3resource' Expected output: {'w': 1, '3': 1, 'r': 2, 'e': 2, 's': 1, 'o': 1, 'u': 1, 'c': 1}

Program

```
string = "w3resource"
count_dict = {}
for char in string:
    if char in count_dict:
        count_dict[char] += 1
    else:
        count_dict[char] = 1
print(count_dict)
```

Output:

```
{'w': 1, '3': 1, 'r': 2, 'e': 2, 's': 1, 'o': 1, 'u': 1, 'c': 1}
```

Exercise: 7

Difficulty: Hard

Question: Write a Python program to take employee records (ID, name, salary) in a dictionary and print the top 3 highest-paid employees.

Program

```
# employees = {}
n = int(input("Enter number of employees: "))
for i in range(n):
    emp_id = input("Enter Employee ID: ")
    name = input("Enter Employee Name: ")
    salary = float(input("Enter Employee Salary: "))
    employees[emp_id] = (name, salary)
sorted_employees = sorted(employees.items(), key=lambda x: x[1][1], reverse=True)
print("\nTop 3 Highest-Paid Employees:")
for emp_id, (name, salary) in sorted_employees[:3]:
    print(f"ID: {emp_id}, Name: {name}, Salary: {salary}")
```

Output:

```
Enter number of employees: 3
Enter Employee ID: 1706
Enter Employee Name: Rudresh
Enter Employee Salary: 100000
Enter Employee ID: 1712
Enter Employee Name: Aakash
Enter Employee Salary: 90000
Enter Employee ID: 1704
Enter Employee Name: Dhivas
Enter Employee Salary: 950000

Top 3 Highest-Paid Employees:
ID: 1704, Name: Dhivas, Salary: 950000.0
ID: 1706, Name: Rudresh, Salary: 100000.0
ID: 1712, Name: Aakash, Salary: 90000.0
```

Assignment Evaluation:

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Viva Questions

1. What is a dictionary in Python and how is it defined?
2. How do dictionaries differ from lists and tuples?
3. What are keys and values in a dictionary?
4. Can a dictionary have duplicate keys? Justify your answer.
5. How do you access the value associated with a specific key?

